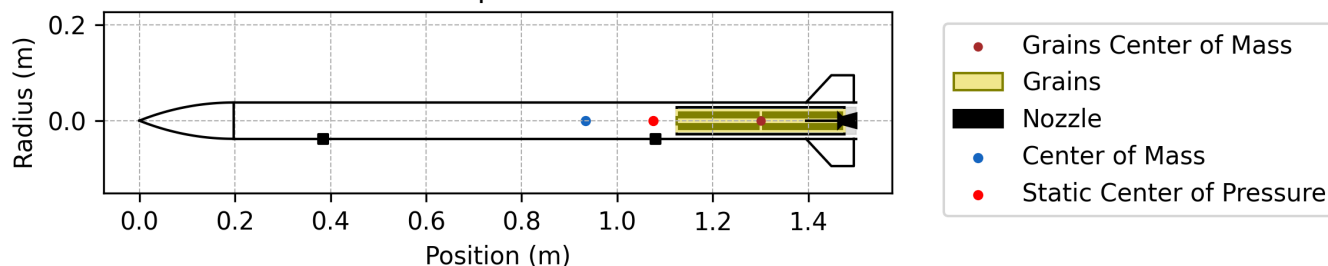
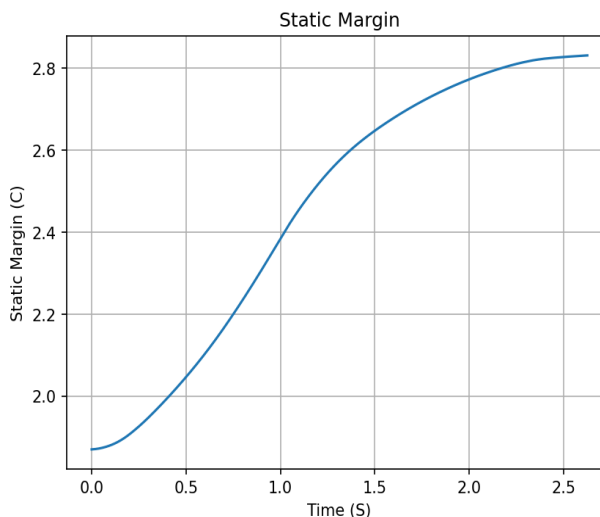
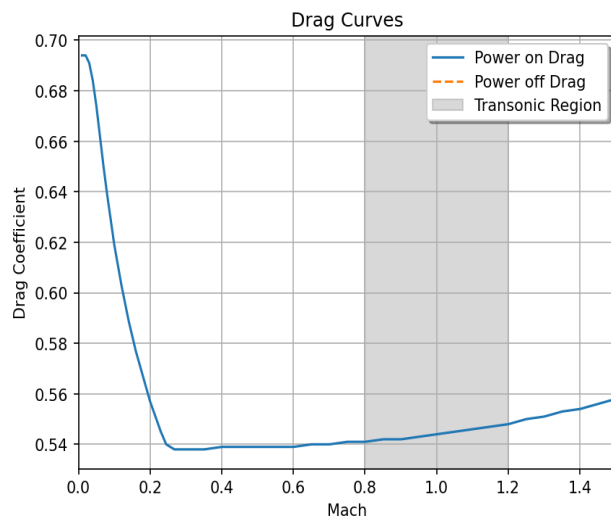


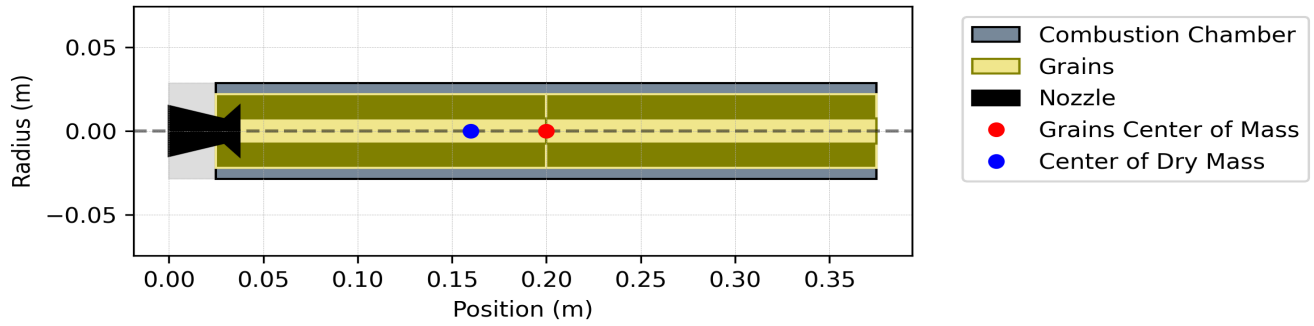
Rocket Representation



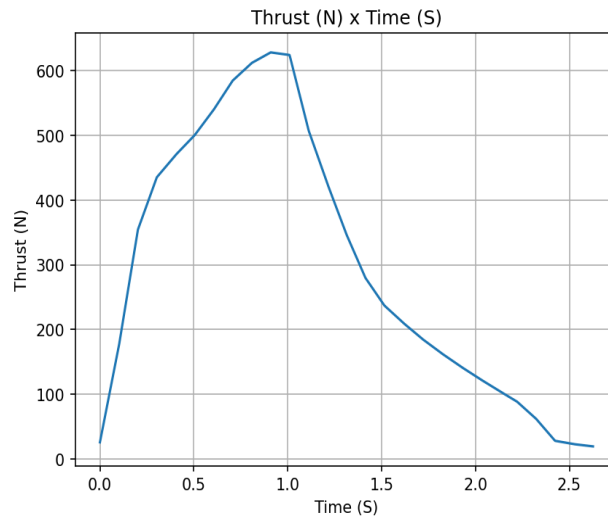
Rocket Parameters			
Maximum Radius [m]	0.038	CoDM - CoM (no motor) [m]	0.125
Rocket Mass (without motor) [kg]	3.275	CoDM - Nozzle Exit [m]	0.640
Rocket Dry Mass (with unloaded motor) [kg]	4.131	CoDM - CoPM (t=0) [m]	0.440
Rocket Loaded Mass (t=0) [kg]	4.954	CoM (t=0) - CoDM [m]	0.073
Stability			
Initial Static Margin (mach=0, time=0)	1.870 c	Final Static Margin (mach=0, time=burn_out)	2.831 c
Dry Inertias			
Inertia I11 dry [kg·m ²]	0.723	Inertia I12 dry [kg·m ²]	0.000
Inertia I22 dry [kg·m ²]	0.723	Inertia I13 dry [kg·m ²]	0.000
Inertia I33 dry [kg·m ²]	0.002	Inertia I23 dry [kg·m ²]	0.000
Filled Inertias			
Filled Inertia I11 [kg·m ²]	0.890	Filled Inertia I12 [kg·m ²]	0.000
Filled Inertia I22 [kg·m ²]	0.890	Filled Inertia I13 [kg·m ²]	0.000
Filled Inertia I33 [kg·m ²]	0.003	Filled Inertia I23 [kg·m ²]	0.000
Parachutes			
Parachute count	1		
Main Parachute CdS [m ²]	N/A	Main Parachute Sampling Rate [Hz]	N/A
Main Parachute Lag Time [s]	N/A	Main Parachute Trigger	N/A
Drogue Parachute CdS [m ²]	1.5	Drogue Parachute Sampling Rate [Hz]	105.0
Drogue Parachute Lag Time [s]	1.5	Drogue Parachute Trigger	750 m (AGL)



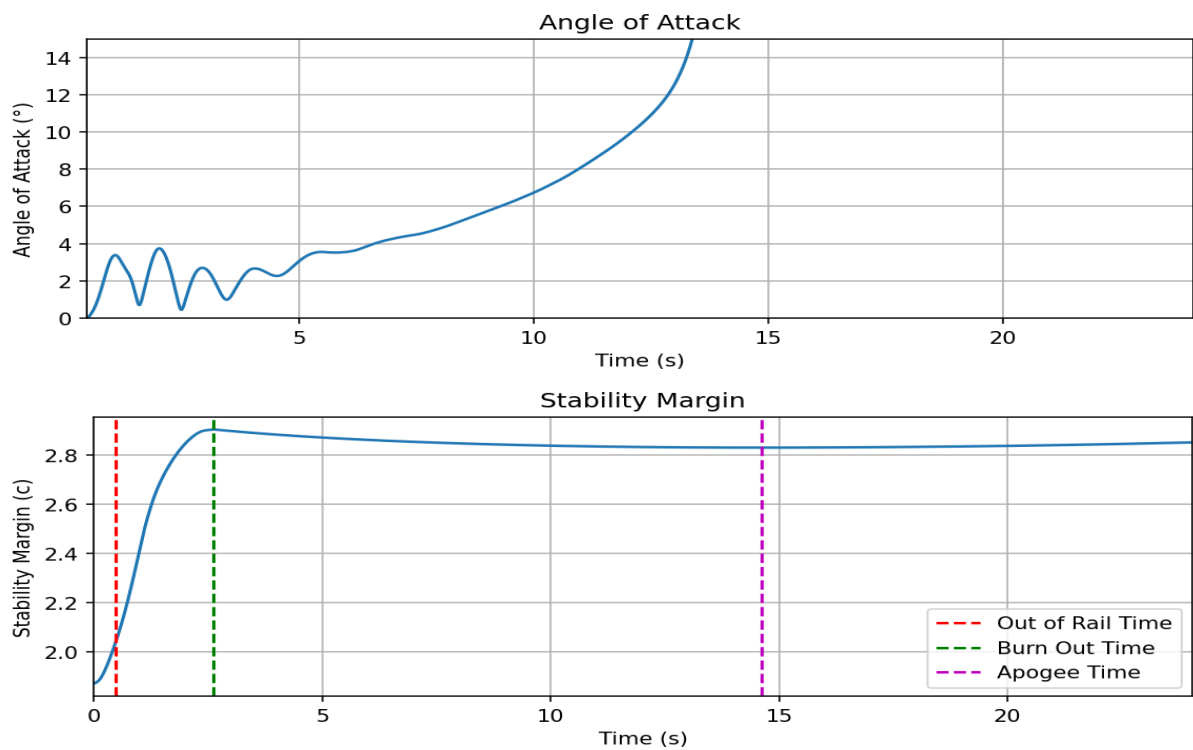
Solid Motor Representation



Motor Parameters			
Total Burning Time [s]	2.626	Total Impulse [Ns]	795.395
Total Structural Mass [kg]	0.856	Total Propellant Mass [kg]	0.823
Total Initial Mass (t=0) [kg]	1.679		
Dry Inertias			
Dry I11 [kg·m ²]	0.000	Dry I12 [kg·m ²]	0.000
Dry I22 [kg·m ²]	0.000	Dry I13 [kg·m ²]	0.000
Dry I33 [kg·m ²]	0.000	Dry I23 [kg·m ²]	0.000
Grain Details			
Number of Grains	2	Grain Spacing [m]	0
Grain Density [kg/m ³]	1750	Grain Outer Radius [m]	0.022
Grain Inner Radius [m]	0.0075	Grain Height [m]	0.175
Grain Volume [m ³]	0.000 (20 dp)	Grain Mass [kg]	0.412

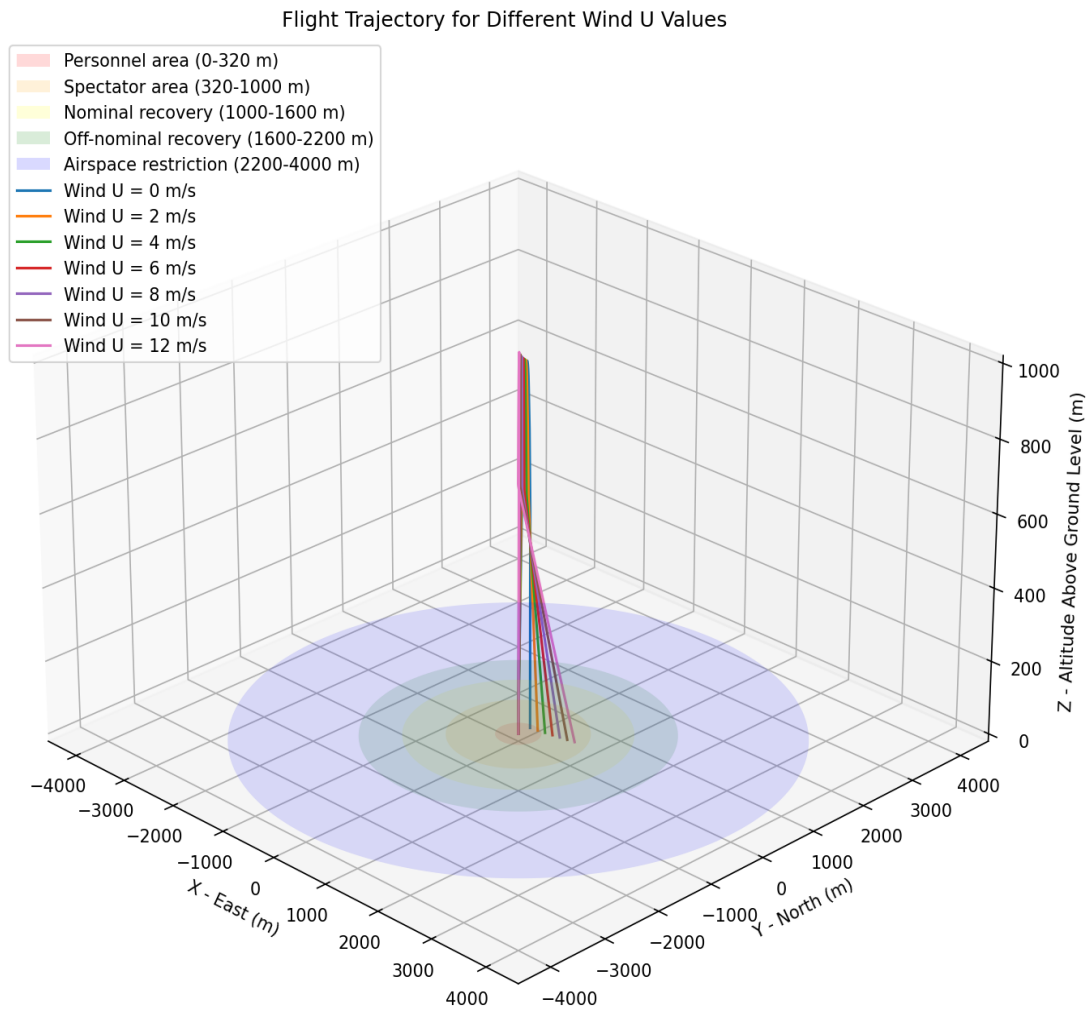
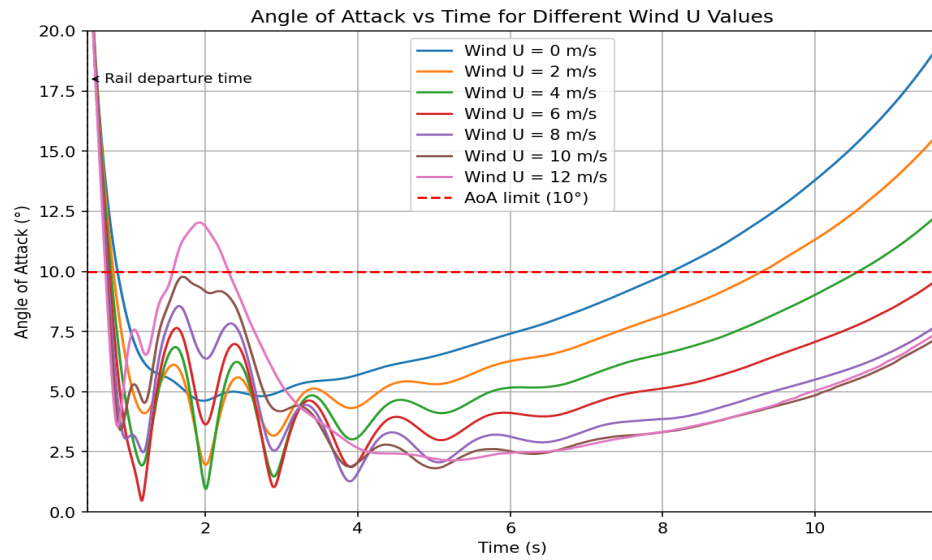


Flight Parameters			
Initial Stability Margin [c]	1.870	Rail Departure Time [s]	0.483
Rail Departure Velocity [m/s]	28.234	Rail Departure Stability Margin [c]	2.040
Rail Departure AoA [°]	0.000		
Burn out State			
Alt at burn out (ASL)	715.611	Alt at burn out (AGL)	245.611
Speed at burn out [m/s]	140.526	Freestream vel at burn out [m/s]	140.526
Mach at burn out	0.417		
Apogee State			
Apogee Time [s]	14.621	Apogee Alt (ASL) [m]	1480.663
Apogee Alt (AGL) [m]	1010.663	Apogee Freestream Speed [m/s]	7.507
Apogee Drift [m]	114.429		
Impact / End of Simulation			
Impact Time [s]	114.658	Impact Drift [m]	477.482
Vertical vel at impact [m/s]	-6.894	# parachutes until impact	1
Maximum Values			
Maximum Speed [m/s]	143.294	At time [s]	2.30
Maximum Mach	0.425	At time [s]	2.30
Stability Margin Summary			
Initial Stability Margin [c]	1.870	At time [s]	0.00
Out of Rail Stability Margin [c]	2.040	At time [s]	0.48
Minimum Stability Margin [c]	1.870	At time [s]	0.00
Parachute Events			
Drogue			
Ejection: 22.552 s Inflation: 24.052 s Speed: 76.083 m/s Altitude: 1113.975 m ASL / 643.975 m AGL			



Flight dynamics safety analysis

This section compares several flights under varying magnitudes of winds heading East, evaluating flight dynamics safety: angle-of-attack evolution, 3D trajectories with recovery/drift zones. Each figure is shown below.



Impact drift per Wind U: 0 m/s -> 235.8 m; 2 m/s -> 285.3 m; 4 m/s -> 370.3 m; 6 m/s -> 472.6 m; 8 m/s -> 580.1 m; 10 m/s -> 695.7 m; 12 m/s -> 810.0 m.

Highlights Legend

- Red: value outside expected/critical range
- Yellow: attention (non-fatal / unusual)

Highlight explanations

1. [Rocket] Parachute count: Parachute count == 1 (value=1) -> highlighted yellow
2. [Motor] Inertias: Inertia anomaly detected ($I_{33} > I_{11}/I_{22}$ or values out of expected range) -> highlighted red